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Digital Industrial Policy through Data Security? – China’s Approach and Security Exceptions under Trade Agreements

Xiaomeng Qu^{*} & Weihuan Zhou^{**}

ABSTRACT

As industrial policy and national security become increasingly integrated, the extent to which national security may be abused to foster select economic sectors requires more dedicated studies. We make two major contributions in this paper. First, we offer a critical analysis of China’s digital industrial policy focusing on data security regulation and practices. We show that China has sought to maintain a balanced approach towards data liberalization and protection without compelling evidence to suggest abuses of national security to bolster its data sector. Yet, major loopholes remain in China’s regulatory framework leaving room for discretion and abuse in deploying data security measures. As geopolitical tensions and strategic rivalries intensify, a data trade war in the name of national security is not a remote possibility. Second, we argue that the security exceptions under existing trade or digital economy agreements can hardly strike a desirable balance between data liberalization and security. Governments should join forces to modernize these exceptions, ideally via multilateral venues, by deliberating on data security goals and regulatory practices to build transparency and trust and lay the groundwork for negotiating more detailed exceptions for data security.

Keywords: WTO; FTAs; Digital economy agreements; China; Industrial policy; Data flows; Data localization; National security.

I. INTRODUCTION

Industrial policy is not new. For decades, governments have deployed policies to foster the development of selected economic sectors or entities through a wide array of instruments.¹ Yet, no consensus exists in relation to the rationale and efficacy of industrial policies. For example, leading economists have maintained that industrial policies are legitimate and effective, so the critical question is not whether such policies should be deployed but how to make them better.² A recent report of the Organisation for Economic Co-operation and Development (OECD) also

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¹ See generally Dani Rodrik, ‘Industrial Policy for the Twenty-First Century’, September 2004, 1-56; Todd Tucker, ‘Industrial Policy and Planning: What it is and How to Do It Better’, Roosevelt Institute, July 2019, 1-49, <https://rooseveltinstitute.org/wp-content/uploads/2020/07/RI-Industrial-Policy-and-Planning-201707.pdf>.

² Réka Juhász, Nathan Lane, and Dani Rodrik, ‘The New Economics of Industrial Policy’, National Bureau of Economic Research Working Paper 31538 (August 2023) 1-48, <https://www.nber.org/papers/w31538>.

advocated for the important role that industrial policy can play in the pursuit of strategic goals.³ In contrast, the classic criticisms of industrial policy remain influential, concerning misallocation of resources, economic inefficiency, and significant spillovers which adversely affect trading partners and generate trade tensions.⁴ Moreover, while the success of leading East Asian and Latin American economies in pursuing industrialization and economic growth has presented a strong narrative in favour of industrial policy,⁵ some have cautioned that empirical evidence to support “an activist government policy” remains limited.⁶

Today, industrial policy is ubiquitous regardless of the ongoing debate over its rationale and utility. As Nobel laureate Professor Michael Spence has observed,

“At a time of rising geopolitical tensions and supply-chain fragmentation – when national-security considerations are shaping economic policy, and the risks of war seem to be intensifying – industrial policy is all but inevitable.”⁷

Indeed, the current development of industrial policy is no longer confined to consideration of economic competitiveness or efficiency but embraces broader strategic, societal, and political goals. Notably, industrial policy that shapes business decisions and practices or creates national champions and global behemoths is on the rise,⁸ although the West has long criticized and fought against such policies by state-led economies like China for concerns about market distortions and unfair trade practices.⁹ Here, empirical studies have shown that industrial policies have been increasingly introduced by advanced economies to foster specific economic sectors or firms which are already powerful players in global markets.¹⁰ In line with rapidly-expanding policy goals,

³ Chiara Criscuolo et al., ‘An Industrial Policy Framework for OECD Countries: Old Debates, New Perspectives’, OECD Policy Papers No. 127 (May 2022), https://www.oecd.org/en/publications/an-industrial-policy-framework-for-oecd-countries_0002217c-en.html.

⁴ See eg. Simon Evenett et al, ‘The Return of Industrial Policy in Data’, IMF Working Paper No. 2024/001 (January 2024), <https://www.imf.org/en/Publications/WP/Issues/2023/12/23/The-Return-of-Industrial-Policy-in-Data-542828>; Douglas Irwin, ‘The Return of Industrial Policy’, IMF Finance & Development Magazine (June 2023), <https://www.imf.org/en/Publications/fandd/issues/2023/06/the-return-of-industrial-policy-douglas-irwin>; IMF, OECD, World Bank and WTO, Subsidies, Trade, and International Cooperation, Analytical Notes No 2022/001 (April 2022), <https://www.imf.org/en/Publications/analytical-notes/Issues/2022/04/22/Subsidies-Trade-and-International-Cooperation-516660>.

⁵ See eg. above n 1, Rodrik, at 15; Tucker, at 13-16.

⁶ Howard Pack and Kamal Saggi, ‘Is There a Case for Industrial Policy? A Critical Survey’, (2006)21(2) *The World Bank Research Observer* 267-297.

⁷ Michael Spence, ‘In Defense of Industrial Policy’, Project Syndicate (5 May 2023), <https://www.project-syndicate.org/commentary/industrial-policy-us-chips-and-science-act-debate-by-michael-spence-2023-05>.

⁸ See eg. Ruchir Agarwal, ‘Industrial Policy and the Growth Strategy Trilemma’, IMF Finance & Development Magazine (September 2023), <https://www.imf.org/en/Publications/fandd/issues/Series/Analytical-Series/industrial-policy-and-the-growth-strategy-trilemma-ruchir-agarwal>; Willy C. Shih, ‘The New Era of Industrial Policy Is Here’, Harvard Business Review (September-October 2023), <https://hbr.org/2023/09/the-new-era-of-industrial-policy-is-here>; Anshu Siripurapu and Noah Berman, ‘Is Industrial Policy Making a Comeback?’, Council on Foreign Relations (18 September 2023), <https://www.cfr.org/backgrounder/industrial-policy-making-comeback>.

⁹ See Henry Gao and Weihuan Zhou, *Between Market Economy and State Capitalism: China’s State-Owned Enterprises and The World Trading System* (Cambridge: Cambridge University Press, 2022) Ch 1-3. For a recent joint critique of China’s economic system and policies by G7 countries, see The White House, ‘G7 Hiroshima Leaders’ Communique’ (20 May 2023), <https://www.whitehouse.gov/briefing-room/statements-releases/2023/05/20/g7-hiroshima-leaders-communique/>.

¹⁰ Réka Juhász et al., ‘The Who, What, When, and How of Industrial Policy: A Text-Based Approach’, Structural Transformation and Economic Growth WP050 (12 January 2023), <https://steg.cepr.org/publications/who-what-when-and-how-industrial-policy-text-based-approach>.

governments have repurposed and expanded their policy tools and practices¹¹ which are more likely to spill across national borders.¹² However, in deploying these policies, governments tend to prioritize their own national interests without paying sufficient attention to the policies' global implications.¹³

The proliferation of modern industrial policies has gone hand in hand with the expansive use of “national security” beyond military/defense-related interests to pursue wide-ranging economic interests.¹⁴ Supply chain safety and technological leadership in strategic and frontier sectors, and economic competitiveness and industrialization in general, are increasingly viewed via a security lens,¹⁵ blurring the boundary between national security and industrial policy. In this context, governments are retreating from the conventional wisdom of free trade and existing commitments to constrain the abuse of trade policies, while increasingly using trade policies to advance domestic industrial and security interests.¹⁶ This ongoing shift towards inward-oriented economic and security policies has exaggerated the longstanding tension between trade liberalization and domestic autonomy.

Against this backdrop, this paper explores so-called ‘digital industrial policy’ by focusing on data-related policies and regulation. As the world navigates the digital age, data regulation has rapidly fragmented. Governments have introduced and continue to experiment with a variety of regulatory approaches in pursuit of diverse and oftentimes competing policy goals. These range from promoting free flows of data for e-commerce, global business connectivity and supply chains to protecting privacy and cybersecurity, fostering national digital capabilities, and pursuing strategic competition for technological supremacy.

To contribute to the already voluminous literature on data policies and regulation at domestic and international levels, this paper focuses on examining the growing interaction between digital industrial policy and national security, using China as a case study. As explained in Section II, digital industrial policies in major economies have been designed increasingly through a security lens. This trend generates concerns about the extent to which national security may be abused to protect or advance digital industries at home at the cost of foreign competitors. Section III conducts a detailed study on the evolution of China’s digital industrial policy with a focus on data policies and regulation. After a critical analysis of China’s data security laws and practices, particularly those relating to cross-border data transfers and data localization, we find no compelling evidence to suggest that China has used security-based measures to bolster its data sector. Rather, we show that China has sought to maintain a balanced approach towards data

¹¹ Ibid. See also above n 3, Criscuolo et al.

¹² See above n 8, Shih.

¹³ See the references in FN 4, and above n 8, Siripurapu and Berman. See also Bernard Hoekman, Petros Mavroidis and Douglas Nelson, *Non-Economic Objectives, Globalisation and Multilateral Trade Cooperation* (Paris & London: CEPR Press, 2023), <https://cepr.org/publications/books-and-reports/non-economic-objectives-globalisation-and-multilateral-trade>.

¹⁴ See generally H.G. Cohen, ‘Nations and Markets’, (2020)23(4) *Journal of International Economic Law* 793.

¹⁵ See Weihuan Zhou, et al., ‘Trade vs. Security: Recent Developments of Global Trade Rules and China’s Policy and Regulatory Responses from Defensive to Proactive’, (2023)22(2) *World Trade Review* 193, 193-94.

¹⁶ U.S. economic and security policies are driving this paradigm shift and reshaping the global economic order. See e.g. Marc Fasteau and Ian Fletcher, *Industrial Policy for the United States: Winning the Competition for Good Jobs and High-Value Industries* (Cambridge: Cambridge University Press, 2024). For a summary of the latest round of U.S. tariffs and bilateral agreements with major partners for economic and security goals, see The White House, ‘Further Modifying the Reciprocal Tariff Rates’, Executive Order (31 July 2025), <https://www.whitehouse.gov/presidential-actions/2025/07/further-modifying-the-reciprocal-tariff-rates/>.

security by ensuring that data restrictive measures target genuine security risks and do not cause unnecessary burdens on business operations. However, the existence of major ambiguities in China's data security laws, the well-known lack of transparency in China's regulatory practices, etc. may provide room for abuse of national security as industrial policy. To address the systemic challenges posed by the increasingly blurry line between (digital) industrial policy and national security, Section IV discusses the inadequacies of security exceptions in existing trade or digital economy agreements in dealing with data security, focusing on the Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP), the United States–Mexico–Canada Agreement (USMCA), the Digital Economy Partnership Agreement (DEPA), the Regional Comprehensive Economic Partnership (RCEP) and the Joint Statement Initiative on E-commerce (JSI) under the World Trade Organization (WTO). We then propose ways to facilitate negotiations of data security issues with an aim to modernizing the security exceptions to strike a proper balance between data liberalization and security. Section V sets forth concluding remarks.

II. DIGITAL INDUSTRIAL POLICY VIA A SECURITY LENS

As reported by Evenett and Fritz, “[p]olicymakers are flying blind as they shape and nurture the digital domain”, with a better part of this fast-evolving, global regulatory developments driven primarily by fostering selected, domestic digital sectors.¹⁷ While governments have different policy priorities, a common area of regulatory effort has been data¹⁸ due to the pivotal role data plays in digital transformation. Various forms of regulatory interventions have been introduced to build data-related capabilities such as subsidizing R&D, digital start-ups and the establishment of data infrastructures, and enhancing enabling regulatory frameworks for data-based digital businesses.¹⁹ Such interventions have also been used to foster the competitiveness of national tech companies through mobilizing local data collection, restricting cross-border data flows, mandating data localization, promoting outbound foreign investment, etc.²⁰ More broadly, the growing claim for data sovereignty can potentially become an all-encompassing policy that enables the deployment of industrial policy for a diverse range of economic and non-economic goals.²¹

Our key task in this paper is exploring the intersection of national security and digital industrial policy, especially in data regulation. Recent studies have expounded the contrasting models that

¹⁷ Simon Evenett and Johannes Fritz, *Emergent Digital Fragmentation: The Perils of Unilateralism* (London: CEPR Press, 2022) at 5, 14-15, <https://www.hinrichfoundation.com/research/wp/digital/emergent-digital-fragmentation-the-perils-of-unilateralism/>.

¹⁸ Ibid., at 21.

¹⁹ See generally Parminder Jeet Singh, ‘Digital Industrialisation in Developing Countries – A Review of the Business and Policy Landscape’, IT for Change Research Paper (December 2017), <https://itforchange.net/sites/default/files/1468/Digital-industrialisation-May-2018.pdf>.

²⁰ Ibid. For a comprehensive discussion of digital policies and transformation in the US, China and the EU, see generally Anu Bradford, *Digital Empires: The Global Battle to Regulate Technology* (New York: Oxford University Press, 2023).

²¹ See generally Melody Musoni et al., ‘Global Approaches to Digital Sovereignty: Competing Definitions and Contrasting Policy’, ECDPM Discussion Paper No. 344 (May 2023), <https://ecdpm.org/application/files/7816/8485/0476/Global-approaches-digital-sovereignty-competing-definitions-contrasting-policy-ECDPM-Discussion-Paper-344-2023.pdf>; Anupam Chander and Haochen Sun, ‘Sovereignty 2.0’ in Anupam Chander & Haochen Sun (eds) *Data Sovereignty: From the Digital Silk Road to the Return of the State* (New York: Oxford University Press, 2023) ch 1.

major economies have adopted in digital/data regulation, with the U.S. model focused on the protection of firms, the EU model on the protection of individuals, and the Chinese model on the protection of the state.²² While this distinction captures the essence of the respective regulatory priorities, it does not offer sufficient insight into the intricacies associated with the growing intersection of economic and security matters, including in the digital domain.

There is already abundant evidence to show strengthened regulatory linkages between digital industrial policy and national security. One major example concerns the surge of U.S. industrial policy in recent times to promote U.S. global competitiveness in critical sectors, technologies, and manufacturing capabilities.²³ These industrial policies, officially labelled as the “modern American industrial strategy”, have been developing noticeably through a security lens which may involve all matters related to economic prosperity and opportunity, democratic values, U.S. global leadership, and strategic and geopolitical competition with China.²⁴ To remain the world leader in critical and frontier technologies, the U.S. industrial strategy is determined to promote advanced manufacturing where data is the backbone of many targeted sectors.²⁵ The Biden administration already constrained outbound data transfers to certain countries of concern which may use the data to develop strategic advantages over the U.S., threatening U.S. national security and foreign policy.²⁶ While this restriction on data flows to selected countries is apparently driven by security concerns, the US government recognizes that data protection can serve multiple goals such as protecting U.S. leadership and competitiveness.²⁷ The CHIPS and Science Act of 2022, for instance, adopted a subsidy-based industrial policy to advance U.S. global leadership in

²² See generally above n 20, Bradford; Henry Gao, ‘Digital or Trade? The Contrasting Approaches of China and US to Digital Trade’, (2018)21(2) *Journal of International Economic Law* 297; Henry Gao, ‘Data Sovereignty and Trade Agreements: Three Digital Kingdoms’ in Anupam Chander & Haochen Sun (eds) *Data Sovereignty: From the Digital Silk Road to the Return of the State* (New York: Oxford University Press, 2023) ch 9.

²³ See generally Scott Lincicome, ‘Made in America: The Boom in U.S. Manufacturing Investment’, Testimony to the Joint Economic Committee of the United States Congress (12 June 2024), <https://www.cato.org/testimony/made-america-boom-us-manufacturing-investment>; above n 8, Siripurapu and Berman.

²⁴ See generally The White House, *National Security Strategy* (October 2022), <https://www.whitehouse.gov/wp-content/uploads/2022/11/8-November-Combined-PDF-for-Upload.pdf>; The White House, ‘Remarks on Executing a Modern American Industrial Strategy by NEC Director Brian Deese’ (13 October 2022), <https://www.whitehouse.gov/briefing-room/speeches-remarks/2022/10/13/remarks-on-executing-a-modern-american-industrial-strategy-by-nec-director-brian-deese/>; The White House, ‘Remarks by National Security Advisor Jake Sullivan on Renewing American Economic Leadership at the Brookings Institution’ (27 April 2023), <https://www.whitehouse.gov/briefing-room/speeches-remarks/2023/04/27/remarks-by-national-security-advisor-jake-sullivan-on-renewing-american-economic-leadership-at-the-brookings-institution/>; U.S. Department of the Treasury, ‘Remarks by Secretary of the Treasury Janet L. Yellen on the U.S. - China Economic Relationship at Johns Hopkins School of Advanced International Studies’ (20 April 2023), <https://home.treasury.gov/news/press-releases/jv1425>; Andres B. Schwarzenberg, ‘Industrial Policy and International Trade’, U.S. Congressional Research Service (9 August 2024), <https://crsreports.congress.gov/product/pdf/IF/IF12119>.

²⁵ See generally National Science and Technology Council, *National Strategy for Advanced Manufacturing*, Report by the Subcommittee on Advanced Manufacturing and Committee on Technology (October 2022), <https://www.whitehouse.gov/wp-content/uploads/2022/10/National-Strategy-for-Advanced-Manufacturing-10072022.pdf>.

²⁶ The White House, ‘Executive Order on Preventing Access to Americans’ Bulk Sensitive Personal Data and United States Government-Related Data by Countries of Concern’ (28 February 2024), <https://www.whitehouse.gov/briefing-room/presidential-actions/2024/02/28/executive-order-on-preventing-access-to-americans-bulk-sensitive-personal-data-and-united-states-government-related-data-by-countries-of-concern/>.

²⁷ See above n 25, National Science and Technology Council, at 8.

semiconductors, technologies of the future and the relevant supply chains, treating this leadership as being crucial for protecting U.S. national security.²⁸ Another case in point concerns the regulatory development on artificial intelligence (AI). President Biden committed unwaveringly to advancing U.S. global leadership in AI technologies, infrastructure and capabilities as a matter of national security significance.²⁹ While abandoning Biden’s AI policies, President Trump became even more committed to “enhance[ing] America’s global AI dominance to promote ... economic competitiveness, and national security.”³⁰ Given the importance of data for the training of AI, the recent development of U.S. AI policies sends a signal that more regulation may be introduced to tighten control over data.³¹

Similar regulatory developments where industrial policy and national security become increasingly integrated are also underway in other major economies. For example, faced with geopolitical shocks and uncertainties, the EU’s “New Industrial Strategy” becomes inevitably multi-purpose with an emphasis on strengthening its “strategic autonomy”.³² When it comes to the digital domain, the Digital Agenda for Europe is currently focused on fulfilling the EU’s technological and geopolitical goals by prioritizing critical and emerging technologies such as semiconductors, quantum computing as well as security-related interests such as cybersecurity and digital sovereignty.³³ As such, the EU’s digital industrial policy treats digital capabilities and competitiveness as essential security interests.³⁴ The EU’s data strategy in particular, while

²⁸ The White House, ‘FACT SHEET: CHIPS and Science Act Will Lower Costs, Create Jobs, Strengthen Supply Chains, and Counter China’ (9 August 2022), <https://www.whitehouse.gov/briefing-room/statements-releases/2022/08/09/fact-sheet-chips-and-science-act-will-lower-costs-create-jobs-strengthen-supply-chains-and-counter-china/>; The White House, ‘FACT SHEET: Two Years after the CHIPS and Science Act, Biden-Harris Administration Celebrates Historic Achievements in Bringing Semiconductor Supply Chains Home, Creating Jobs, Supporting Innovation, and Protecting National Security’ (9 August 2024), <https://www.whitehouse.gov/briefing-room/statements-releases/2024/08/09/fact-sheet-two-years-after-the-chips-and-science-act-biden-%e2%81%a0harris-administration-celebrates-historic-achievements-in-bringing-semiconductor-supply-chains-home-creating-jobs-supporting-inn/>.

²⁹ The White House, ‘Memorandum on Advancing the United States’ Leadership in Artificial Intelligence; Harnessing Artificial Intelligence to Fulfill National Security Objectives; and Fostering the Safety, Security, and Trustworthiness of Artificial Intelligence’ (24 October 2024), <https://www.whitehouse.gov/briefing-room/presidential-actions/2024/10/24/memorandum-on-advancing-the-united-states-leadership-in-artificial-intelligence-harnessing-artificial-intelligence-to-fulfill-national-security-objectives-and-fostering-the-safety-security/>.

³⁰ The White House, ‘Fact Sheet: President Donald J. Trump Takes Action to Enhance America’s AI Leadership’ (23 January 2025), <https://www.whitehouse.gov/fact-sheets/2025/01/fact-sheet-president-donald-j-trump-takes-action-to-enhance-americas-ai-leadership/>.

³¹ For an interesting discussion of how the development of AI policies may increase the pursuit of digital/data sovereignty via industrial policy, see generally Andrew Keane Woods, ‘Digital Sovereignty + Artificial Intelligence’ in Anupam Chander & Haochen Sun (eds) *Data Sovereignty: From the Digital Silk Road to the Return of the State* (New York: Oxford University Press, 2023) ch 5.

³² See generally Simone Tagliapietra and Reinhilde Veugelers, ‘Industrial Policy in Europe: Past and Future’ in Simone Tagliapietra and Reinhilde Veugelers (eds) *Sparkling Europe’s New Industrial Revolution: A Policy for Net Zero, Growth and Resilience* (Belgium: Bruegel Blueprint Series, 2023) ch 1; The European Parliament, ‘General Principles of EU Industrial Policy’, Fact Sheets on the European Union (undated), <https://www.europarl.europa.eu/factsheets/en/sheet/61/general-principles-of-eu-industrial-policy>.

³³ The European Parliament, ‘Digital Agenda for Europe’, Fact Sheets on the European Union (undated), <https://www.europarl.europa.eu/factsheets/en/sheet/64/digital-agenda-for-europe>.

³⁴ The European Commission, ‘European Industrial Strategy’ (undated), https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/europe-fit-digital-age/european-industrial-strategy_en; The European Commission, ‘Communication from the Commission to the European Parliament, the European Council, the Council, the European Economic and Social Committee and the Committee of the Regions’, COM(2020)102 final (10 March 2020), <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52020DC0102>.

maintaining a focus on the protection of individuals, sets forth a comprehensive blueprint for enhancing EU's data-related technologies, infrastructures and capabilities as a crucial way to safeguard its fundamental values, rights and digital sovereignty.³⁵

Digital industrial policy has also gained popularity in developing economies. While these policies are aimed at advancing domestic digital sectors including the data industry, they are also driven by non-economic goals particularly national security and (data) sovereignty.³⁶ Consequently, policy tools, such as data blocking and localization, are put in place to serve multiple objectives, demonstrating growing linkages between industrial policy and national security.³⁷ India has presented a more proactive case given the comprehensiveness of India's digital industrial policy and its accomplishments in enabling the regulatory environment and mobilizing diverse resources to foster the development of its digital firms and data-related infrastructures and capabilities.³⁸ At the same time, India's digital industrialization has relied on a range of laws and regulations to protect digital security currently focusing on data security and sovereignty.³⁹ Data control measures such as localization requirements are employed not only as protectionist instruments but also as security instruments under India's overarching digital industrial policy.⁴⁰

In short, as the global race for digital transformation intensifies, digital industrial policy has developed to an unprecedented scale underpinned by both economic and security considerations.⁴¹ This emerging trend in the development of digital industrial policy requires more detailed study to understand how such policies now interact with security interests.

III. THE EVOLUTION OF CHINA'S DATA REGULATION: NATIONAL SECURITY AS INDUSTRIAL POLICY?

China offers a good case study for at least two reasons: (1) China's economic growth has relied heavily on industrial policies for decades, and (2) the ambitious and fast-developing digital industrial policy in China has prioritized economic and security goals. Through a careful review and analysis of China's data-related policies and regulation, we seek to expound the extent to which the development of these policies has been shaped by security concerns and how national

³⁵ The European Commission, 'Communication from the Commission to the European Parliament, the European Council, the Council, the European Economic and Social Committee and the Committee of the Regions', COM(2020)66 final (19 February 2020), <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52020DC0066>; Macmillan Keck, 'The Role of Cross-Border Data Flows in the Digital Economy', UNCDF (Jul. 2022) at 11-12, <https://policyaccelerator.uncdf.org/all/brief-cross-border-data-flows#:~:text=In%20a%20digital%20economy%2C%20cross,transfer%20of%20goods%20or%20services>.

³⁶ Christopher Foster and Shamel Azmeh, 'Latecomer Economies and National Digital Policy: An Industrial Policy Perspective', (2020)56(7) *Journal of Development Studies* 1247, 1257; Sanghyun Han, 'Data and Statecraft: Why and How States Localize Data', (2024)26(2) *Business and Politics* 263, 273-80.

³⁷ Ibid.

³⁸ See above n 19, Singh, at 35-56. For a more comprehensive discussion of India's digital strategy, see Li Zhang and Dayi Hu, 'National Digital Development Strategy and Its Practice in India' in Guang Yang et al. (eds) *Countries and Regions: Dynamic Interconnectivity* (Singapore: Palgrave Macmillan, 2024) 137-181.

³⁹ International Trade Administration of India, 'Digital Economy' (updated 18 September 2024), <https://www.trade.gov/country-commercial-guides/india-digital-economy>; above n 38, Zhang and Hu, at 165.

⁴⁰ Kaushambi Bagchi et al., 'Data Flows and Data Localisation: An Economic Analysis', Indian Council for Research on International Economic Relations (June 2020) ii-iv, https://icrier.org/pdf/Data_Flows_and_Data_Localisation.pdf; Rudra Chaudhuri and Arjun Kang Joseph, 'Living in a Fragmented World: India's Data Way', (2024)23(2) *India Review* 154, 156-57.

⁴¹ See above n 17, Evenett and Fritz.

security may be (ab)used as a disguised protectionist tool to advance China's global leadership in digital industrialization. For this purpose, we adopt so-called process-tracing methodology by tracing the evolution of China's data regulatory regime based on primary sources.⁴²

1. *China's Digital Industrial Policy in a Nutshell*

A brief review of the evolution of China's overarching digital industrial policy provides an important context for understanding its regulatory framework for data. This evolution dates to the 1990s and covers at least three main phases. In the first phase up to 2005, China's primary goal was to foster the infant information industry, particularly manufacturing capabilities of electronics and information products.⁴³ A well-known example was the growth of the semiconductor sector based on a wide array of subsidies and regulatory support, such as R&D and other grants, tax preferences, export promotion, and government procurement.⁴⁴ More broadly, through similar supportive schemes at both central and local levels, the information industry developed rapidly to become a key pillar of the national economy by 2005.⁴⁵

The second phase (2006-2015) witnessed a shift of policy priorities from building an infant domestic industry to developing China's indigenous innovation and competitiveness in critical technologies.⁴⁶ In light of its technology-and-innovation-oriented development model, China deployed massive resources to advance R&D and innovation in frontier information

⁴² This methodology has proven effective and reliable in demystifying complex regulatory frameworks in the digital space. See eg. Rogier Creemers, 'China's Emerging Data Protection Framework', (2022)8(1) *Journal of Cybersecurity* 1; Han, above n 36.

⁴³ 《中华人民共和国国民经济和社会发展十年规划和第八个五年计划纲要》[Ten-Year Plan and the Eighth Five-Year Plan for Economic and Social Development of the People's Republic of China (1991-1995)], issued in April 1991, <https://www.ndrc.gov.cn/fggz/fzzlgh/gifzgh/200709/P020191029595681819982.pdf>; 《中华人民共和国国民经济和社会发展“九五”计划和 2010 年远景目标纲要》[Ninth Five-Year Plan for Economic and Social Development of the People's Republic of China and the Outlines of Objectives in Perspective of the Year 2010 (1996-2000)], issued on 17 March 1996, <https://www.ndrc.gov.cn/fggz/fzzlgh/gifzgh/200709/P020191029595686994247.pdf>; Feitao Jiang et al., 'China's Industrial Policy in the Digital Economy Era' [数字经济时代的中国产业政策], (2024) 6 *Study& Exploration* 168, 169-174.

⁴⁴ 《国务院关于印发鼓励软件产业和集成电路产业发展若干政策的通知》[Several Policies to Encourage the Development of the Software Industry and Integrated Circuit Industry], issued by the State Council on 24 June 2000, https://www.gov.cn/gongbao/content/2000/content_60310.htm; Alex He, 'China's Techno-Industrial Development: A Case Study of the Semiconductor Industry' (CIGI Papers No.252, May 2021) at 15, <https://www.cigionline.org/static/documents/documents/no.252%20web.pdf>.

⁴⁵ 《中华人民共和国国民经济和社会发展第十个五年计划纲要》[Tenth Five-Year Plan for Economic and Social Development of the People's Republic of China (2001-2005)], issued on 5 March 2001, www.gov.cn/gongbao/content/2001/content_60699.htm; “十五”期间信息化建设成效显著' [Achievements in Informatization Development during the Tenth Five-Year Period] (1 January 2006), https://www.gov.cn/ztl/2006-01/01/content_145195.htm.

⁴⁶ 《2006-2020年国家信息化发展战略》[The National Informatization Development Strategy for 2006-2020], issued by the General Office of the Central Committee of Communist Party of China and the General Office of the State Council on 19 March 2006, https://www.gov.cn/gongbao/content/2006/content_315999.htm.

technologies⁴⁷ and develop supportive digital infrastructures.⁴⁸ These policies enabled China to possess the world's largest broadband network infrastructure within a short period as well as the world's largest internet user base which produces a tremendous amount of data.⁴⁹ In 2015 the Chinese government recognized data as a "basic strategic resource" and since then has placed growing emphasis on the development and application of data in traditional and emerging industries to advance digital transformation.⁵⁰ At the same time, however, the rapid and widespread adoption of digital technologies, such as the internet, generated significant security-related challenges. Since the early 2010s, the Chinese government has taken steps to strengthen security safeguard capabilities by enhancing digital security infrastructure,⁵¹ tightening data protection,⁵² amongst other means. As China's overarching economic policies continued to prioritize development and growth, data regulation in this phase remained piecemeal largely confined to consumer protection.⁵³ Nevertheless, these efforts laid the groundwork for the development of a comprehensive regulatory framework to address security concerns in the following decade.

Balancing development and security interests has been at the centre of China's digital industrial policy in the third phase (2016-current). When it comes to data, the Thirteenth Five-Year Plan (2016-2020) reaffirmed the vital role of data in digital transformation and stressed the need for the openness and sharing of government data, the development of infrastructure such

⁴⁷ 《信息产业“十一五”规划》[Eleventh Five-Year Plan for Information Industry], issued by the Ministry of Industry and Information Technology and Information on 1 March 2007, <https://www.ndrc.gov.cn/fggz/fzzlgh/gjzxgh/200709/P020191104623156010398.pdf>; 《关于印发国家十二五科学技术发展规划的通知》[Development Plan of Science and Technology for the Twelfth Five-Year Plan], issued by the Ministry of Science and Technology on 4 July 2011, https://www.most.gov.cn/xxgk/xinxifenlei/fdzdgnr/qtwj/qtwj2011/201107/t20110713_88228.html.

⁴⁸ 《中华人民共和国国民经济和社会发展第十一个五年规划纲要》[Eleventh Five-Year Plan for Economic and Social Development of the People's Republic of China (2006-2010)], issued on 14 March 2006, https://www.gov.cn/gongbao/content/2006/content_268766.htm; 《中华人民共和国国民经济和社会发展第十二个五年规划纲要》[Twelfth Five-Year Plan for Economic and Social Development of the People's Republic of China (2010-2015)], issued on 14 March 2011, https://www.gov.cn/2011lh/content_1825838.htm.

⁴⁹ ‘“十二五”期间宽带运营发展回顾’[Review of the Development of Broadband Operation Industry during the Twelfth Five-Year Period] (2 August 2017), https://www.ndrc.gov.cn/xwdt/gdzt/xyqqd/201708/t20170802_1197809_ext.html.

⁵⁰ 《国务院关于积极推进“互联网+”行动的指导意见》[Guiding Opinions on Promoting the ‘Internet Plus’ Action Plan], issued by the State Council on 1 July 2015, https://www.gov.cn/zhengce/content/2015-07/04/content_10002.htm; 《国务院关于印发促进大数据发展行动纲要的通知》[Action Outline for Promoting the Development of Big Data], issued by the State Council on 31 August 2015, https://www.gov.cn/zhengce/content/2015-09/05/content_10137.htm.

⁵¹ 《国务院关于大力推进信息化发展和切实保障信息安全的若干意见》[Several Opinions on Promoting the Development of Informatization and Effectively Ensuring Information Security], issued by the State Council on 28 June 2012, effective on 17 July 2012, https://www.gov.cn/zhengce/content/2012-07/17/content_5906.htm. In 2015, the Chinese government implemented 96 special projects for national information security. See 《国家信息安全专项及下一代互联网技术研发、产业化和规模商用专项项目清单》[List of Projects for National Information Security, Next-Generation Internet Technology R&D, Industrialization, and Large-Scale Commercial Application], issued by the National Development and Reform Commission on 16 February 2015, <https://zfxgk.ndrc.gov.cn/web/iteminfo.jsp?id=3331>.

⁵² See above n 51, Several Opinions. For specific regulations, see Kemeng Cai, ‘Jurisdictional Report: The People's Republic of China’ in Clarisse Girot (eds), *Regulation of Cross-border Transfers of Personal Data in Asia*, ABLI Legal Convergence Series (May 2018) at 67-72, <https://abli.asia/abli-publications/regulation-of-cross-border-transfers-of-personal-data-in-asia-softcover/>.

⁵³ See above n 42, Creemers, at 4.

as data platforms and centres, the advancement of data-related technologies, etc.⁵⁴ The Plan also mandated the integration of data security into digital development by establishing a comprehensive regulatory framework for strengthening data protection across the entire data lifecycle.⁵⁵ The first major milestone in this balancing act was the enactment of the Cybersecurity Law (CSL)⁵⁶ in 2016, which became the cornerstone of China’s digital security legislation.⁵⁷ Under the current fourteenth five-year period (2021–2025), the balancing act has continued to evolve through new policy priorities and regulatory interventions.

The “development” side has focused on broadening and deepening digital industrialization in both traditional and emerging sectors (e.g. communication equipment, core electronic components, and critical software vs. AI, big data, blockchain, and cloud computing).⁵⁸ Data is regarded as a “core engine” to drive digital industrialization so that a national data market is needed to establish a comprehensive data-related supply chain involving the R&D, creation, collection, transfer, application, and storage of data.⁵⁹ For instance, new approaches have been introduced to facilitate the opening up of business data by leading digital and internet platforms and to promote cross-industry data interconnectivity and sharing through trusted data spaces collaboratively built by enterprises, research institutions, and industry organizations.⁶⁰ One of the latest developments was the establishment of China’s first state entity specializing in data integration and technology, which builds a platform for data sharing among highways, railways, waterways, aviation and ports, thereby creating a more competitive and innovative servicing ecosystem.⁶¹

⁵⁴ 《中华人民共和国国民经济和社会发展第十三个五年规划纲要》 [Thirteenth Five-Year Plan for National Economic and Social Development of the People’s Republic of China (2016–2020)], issued on 17 March 2016, https://www.gov.cn/xinwen/2016-03/17/content_5054992.htm; 《国务院关于印发“十三五”国家信息化规划的通知》 [National Informatization Plan for the Thirteenth Five-Year Period], issued by the State Council on 15 December 2016, https://www.gov.cn/zhengce/content/2016-12/27/content_5153411.htm.

⁵⁵ Ibid.

⁵⁶ 《中华人民共和国网络安全法》 [Cybersecurity Law of the People’s Republic of China 2016] [hereinafter ‘CSL’], promulgated by the Standing Committee of the National People’s Congress on 7 November 2016, effective on 1 June 2017, http://www.npc.gov.cn/zgrdw/npc/xinwen/2016-11/07/content_2001605.htm.

⁵⁷ See above n 42, Creemers, at 4.

⁵⁸ 《中华人民共和国国民经济和社会发展第十四个五年规划和 2035 年远景目标纲要》 [Outline of the Fourteenth Five-Year Plan for the National Economic and Social Development of the People’s Republic of China and 2035 Long Term Goals (2021–2025)], issued on 12 March 2021, https://www.gov.cn/xinwen/2021-03/13/content_5592681.htm.

⁵⁹ 《国务院关于印发“十四五”数字经济发展规划的通知》 [Digital Economy Development Plan for the Fourteenth Five-year Period], issued by the State Council on 12 December 2021, https://www.gov.cn/zhengce/content/2022-01/12/content_5667817.htm.

⁶⁰ 《关于促进数据产业高质量发展的指导意见》 [Guiding Opinions on Promoting the High-Quality Development of the Data Industry], issued jointly by six State ministries and departments on 28 December 2024, https://www.gov.cn/zhengce/zhengceku/202412/content_6995430.htm. For detailed policies on the opening and sharing of public and enterprise data, see 《关于加快公共数据资源开发利用的意见》 [Opinions on Accelerating the Development and Utilization of Public Data Resources], issued by the General Office of the Central Committee of Communist Party of China and the General Office of the State Council on 21 September 2024, https://www.gov.cn/zhengce/202410/content_6978911.htm; 《关于促进企业数据资源开发利用的意见》 [Opinions on Promoting the Development and Utilization of Enterprise Data Resources], issued by five State ministries and departments on 20 December 2024, https://www.gov.cn/zhengce/zhengceku/202412/content_6994570.htm.

⁶¹ Kandy Wong, ‘How China manages big data is changing, as new state-owned firm takes the helm’, SCMP (20 December 2024), <https://www.scmp.com/economy/china-economy/article/3291731/how-china-manages-big-data-changing-new-state-owned-firm-takes-helm>.

On the “security” side, the Data Security Law⁶² (DSL) and the Personal Information Protection Law⁶³ (PIPL) were enacted at the outset of the five-year period to provide detailed rules in specific areas of digital security, and cumulatively, to create a comprehensive regulatory framework for safeguarding the digital ecosystem. These were followed by the promulgation of a series of implementing regulations detailing the standards and procedures of existing and new security-related measures, such as security review, standard data security contract, and personal information protection certification,⁶⁴ which are further discussed below.

A raft of criticisms have been levelled at China’s digital industrial policy especially the security-related measures. For example, the CSL provoked an intense debate at the WTO with major players such as the US, the EU and Japan criticizing the law for creating protectionist and trade-restrictive instruments in the guise of cybersecurity.⁶⁵ Commentators have similarly criticized China’s security-related data localization requirements and cross-border data flow restrictions as digital protectionism aiming to boost Chinese tech firms at the cost of foreign competitors.⁶⁶ Has China used security-related measures to foster its digital industrialization? To answer this question, a more detailed assessment of China’s balancing act in data regulation is warranted.

2. *Balancing Development and Security in Data Regulation*

As mentioned, the CSL, DSL and PIPL have consolidated and strengthened China’s data protection mechanisms.⁶⁷ To expound China’s balanced approach, we focus on analyzing the security measures related to data localization and cross-border data flows below. In general, the legislation above has established a tiered risk management system for protecting data security based on the importance of data, operators and security interests. This system does not impose a blanket restriction on the cross-border flow or localization of data. Rather, it adopts different

⁶² 《中华人民共和国数据安全法》[Data Security Law of the People’s Republic of China] [hereinafter ‘DSL’], promulgated by the Standing Committee of the National People’s Congress on 10 June 2021, effective on 1 September 2021, http://www.npc.gov.cn/c2/c30834/202106/t20210610_311888.html.

⁶³ 《中华人民共和国个人信息保护法》[Personal Information Protection Law of the People’s Republic of China] [hereinafter ‘PIPL’], promulgated by the Standing Committee of the National People’s Congress on 20 August 2021, effective on 1 November 2021, https://www.gov.cn/xinwen/2021-08/20/content_5632486.htm.

⁶⁴ 《数据出境安全评估办法》[Measures for Security Assessment of Outbound Data Transfer 2022] [hereinafter ‘Security Assessment Measures’], issued by the Cyberspace Administration of China on 7 July 2022, effective on 1 September 2022, https://www.gov.cn/zhengce/zhengceku/2022-07/08/content_5699851.htm; 《个人信息出境标准合同办法》[Measures on Standard Contract for Outbound Transfers of Personal Information] [hereinafter ‘Standard Contract Measures’], issued by the Cyberspace Administration of China on 22 February 2023, effective on 1 June 2023, https://www.gov.cn/zhengce/202311/content_6917770.htm; 《关于实施个人信息保护认证的公告》[Announcement on the Implementation of Personal Information Protection Certification], issued by the State Administration for Market Regulation and the Cyberspace Administration of China on 4 November 2022, https://www.samr.gov.cn/rzjgs/zcfg/art/2022/art_c7cf688ac2944d84b641008252e31db8.html.

⁶⁵ See eg. WTO, Committee on Technical Barriers to Trade, ‘Minutes of the Meeting of 29-30 March 2017’, G/TBT/M/71 (2 June 2017) 3-6.

⁶⁶ Council on Foreign Relations, ‘The Rise of Digital Protectionism’ (18 October 2017), <https://www.cfr.org/report/rise-digital-protectionism>. For detailed discussions of digital protectionism, see eg. Susan Ariel Aaronson, ‘What Are We Talking about When We Talk about Digital Protectionism?’, (2019) 18(4) *World Trade Review* 541.

⁶⁷ The CSL not only unified the fragmented rules on data localization and cross-border transfer scattered in different prior regulations. It also reduced the relevant restrictions by requiring data localization only in certain circumstances and permitting cross-border data transfers when prescribed conditions are met, a more balanced approach subsequently reaffirmed in the DSL and PIPL. See above n 56, CSL, art 37; above n 62, DSL, art 31; above n 63, PIPL, art 40. To compare with the previous restrictions, see above n 52, Cai, at 67-72.

levels of regulatory checks and requirements commensurate to the severity of security risks and impacts.

(a) Data Localization

Data localization is required in three major situations. The first concerns personal information and “important data” collected and generated by critical information infrastructure operators (CIIOs).⁶⁸ As further clarified in the relevant implementing regulations, “critical information infrastructure” refers to “important network infrastructure and information systems in important industries and sectors, including telecommunications, information services, energy, transportation, hydraulic engineering and water utilities, finance, public services, e-government services, national defense science and technology, and others that once damaged or suffer a data leakage, could severely harm national security, the economy, livelihoods, or the public interest.”⁶⁹ “Important data” covers “data that can endanger national security, economic operations, social stability, or public health and safety if manipulated, destroyed, leaked or illegally obtained or used.”⁷⁰ To provide better clarity and certainty, the DSL mandates competent authorities to formulate “important data” catalogues which specify the scope of important data in specific sectors.⁷¹ As stated by President Xi, the localization requirement targets data generated by industries of the highest economic and social importance and likely to attract the most advanced attacks.⁷² Since CIIOs have greater control over data, data localization enables easier monitoring of local servers and faster responses to data leaks and cyberattacks.⁷³

The second situation requires non-CIIOs to localize “important data”.⁷⁴ Thus, this localization requirement targets the importance of data and the interests that such data may affect, rather than who controls the data. For instance, personal and business data held by private firms can be subject to this requirement if the data falls within the “important” category.⁷⁵ As noted above, the development of “important data” catalogues is delegated to competent authorities overseeing specific industries. One example is the automotive industry which possesses increasingly advanced data processing capabilities and generates huge volumes of data.⁷⁶ A joint measure

⁶⁸ See above n 56, CSL, art 37.

⁶⁹ 《关键信息基础设施安全保护条例》 [Regulation on Security Protection of Critical Information Infrastructure], issued by the State Council on 30 July 2021, effective on 1 September 2021, https://www.gov.cn/gongbao/content/2021/content_5636138.htm, art 2.

⁷⁰ See above n 64, Security Assessment Measures, art 19.

⁷¹ See above n 62, DSL, art 21.

⁷² ‘习近平在网络安全和信息化工作座谈会上的讲话’ [Xi Jinping’s Speech at the Symposium on Cybersecurity and Informatization Work], Xinhua News (26 April 2016) https://www.xinhuanet.com/zgjx/2016-04/26/c_135312437_4.htm.

⁷³ Neha Mishra, ‘Privacy, Cybersecurity, and GATS Article XIV: A New Frontier for Trade and Internet Regulation?’ (2020), 19(3) *World Trade Review* 341, 355; Jinhe Liu, ‘China’s Data Localization’, (2020)13(1) *Chinese Journal of Communication*, 84, 91.

⁷⁴ See above n 64, Security Assessment Measures, art 4.1; 《促进和规范数据跨境流动规定》 [Provisions on Facilitating and Standardizing Cross-Border Data Flow 2024] [hereinafter ‘Facilitating and Standardizing Provisions’], issued by the Cyberspace Administration of China on 22 March 2024, https://www.cac.gov.cn/2024-03/22/c_1712776611775634.htm, art 7.2.

⁷⁵ See ‘数据出境安全评估：保护基础性战略资源的重要一环’ [Security Assessment of Outbound Data Transfer: An Important Part of Protecting Basic Strategic Resources], Cyberspace Administration of China (7 August 2017), https://www.cac.gov.cn/2017-08/07/c_1121443948.htm.

⁷⁶ For example, the daily data generation of an autonomous vehicle is estimated at 4,000 gigabytes, which train the car how to behave in real-life road conditions. See Christopher Foster and Shamel Azmeh, ‘Aligning digital and

issued by five central authorities classifies six types of data as being “important”, including geographic data, pedestrian and vehicle flows in sensitive areas (e.g. defense and military administrative zones), data related to certain segments of economic operations (e.g. vehicle flow and logistics), operational data of automobile charging networks, external video and image data (e.g. facial and license information), and other data that may impact national security, public interests and individual rights as decided by the authorities.⁷⁷ Therefore, while “important data” may vary among industries, the localization of such data is driven by concerns about security risks associated with the exposure of sensitive information, leakage of key personnel movements, disclosure of critical national targets (e.g. locations of military and defense industrial bases, and government facilities), theft of R&D data, and damage to critical infrastructure, etc.⁷⁸

The third situation concerns personal information collected and generated by non-CIIOs that process such information where the volume is above prescribed thresholds.⁷⁹ The current thresholds involve a non-CIIO “cumulatively providing personal information of over 1 million individuals (excluding sensitive personal information) or sensitive personal information of over 10,000 individuals to overseas entities since January 1 of the current year.”⁸⁰ While the thresholds specifically target the cross-border transfer of data (see below), they would also trigger the localization requirement.⁸¹ Major technology firms such as Google, Apple, Alibaba hold vast amounts of user information. As explained by a leading Chinese expert and government advisor, if foreign entities obtain such information, they could combine the data with other datasets, using various algorithms for data mining, to extract information that could potentially threaten China’s national security.⁸² Thus, this localization requirement is designed to minimize security risks associated with the transmission of personal data to a degree that may pose security risks due to the rapid development of digital firms and technologies.⁸³

(b) Cross-border Data Flows

Similar to its regulatory approach to data localization, China does not prohibit data holders from transferring data abroad. The tiered risk management mechanism consists of two main layers of requirements: (1) security review, and (2) standard contract or personal data protection certification. The circumstances that would trigger the security review are essentially those in which the localization requirement applies, namely, (1) the transfer of personal information and

industrial policy to foster future industrialization’, Industrial Analytics Platform Insights (May 2023), <https://iap.unido.org/articles/aligning-digital-and-industrial-policy-foster-future-industrialization>.

⁷⁷ 《汽车数据安全管理办法（试行）》 [Provisions on the Management of Automobile Data Security (Trial)], issued by five ministries on 16 August 2021, effective on 1 October 2021, https://www.gov.cn/zhengce/zhengceku/2021-09/12/content_5640023.htm, art 3.

⁷⁸ ‘智能网联汽车信息安全是国家安全的重要防线’ [Information Security of Intelligent Connected Vehicles is a Crucial Defense for National Security] (22 October 2021), <https://www.rmzxw.com.cn/c/2021-10-22/2971170.shtml>.

⁷⁹ See above n 63, PIPL, art 40.

⁸⁰ See above n 74, Facilitating and Standardizing Provisions, art 7.2.

⁸¹ Jihong Chen et al., ‘《数据出境安全评估办法》十四个百分点的理解’ [Understanding the Fourteen Key Issues in the Measures for Security Assessment of Outbound Data Transfer], Zhong Lun Law Firm (8 July 2022), <https://www.zhonglun.com/research/articles/9175.html>.

⁸² Yanqing Hong, ‘The Cross-Border Data Flows Security Assessment: An Important Part of Protecting China’s Basic Strategic Resources’, Yale Law School Paul Tsai China Center Working Paper (20 June 2017), https://law.yale.edu/sites/default/files/area/center/china/document/dataflowssecurity_final.pdf, at 9.

⁸³ Linghan Zhang, ‘个人信息跨境流动制度的三重维度’ [Three Dimensions of the Cross-Border Flow of Personal Information System] (2021) 5 *China Law Review* 37, 41.

“important data” by CIIOs, and (2) the transfer of “important data” or the transfer of personal data beyond the above-mentioned thresholds by non-CIIOs.⁸⁴ According to the Cyberspace Administration of China (CAC), the security review is based on objective criteria to address genuine security risks and balance security concerns with commercial interests by allowing data operators to transfer important data and personal information abroad when necessary for business purposes.⁸⁵ The assessment criteria involve, for example, the necessity and legitimacy of a proposed transfer, the importance of data and the scale of the transfer, the situation of cybersecurity protection and the availability of protective measures for business and personal data in the recipient jurisdiction, etc.⁸⁶ In other circumstances, a security review is not needed before data can be transferred abroad. Where the transfer of data by non-CIIOs is below the prescribed thresholds but involves non-sensitive personal information of over 100,000 individuals, the transferring entity is asked to conclude a standard contract formulated by the CAC with the overseas recipient or obtain a personal information protection certification from a third-party accredited institution according to standards and requirements set forth by the CAC.⁸⁷ Like the security review criteria, the mandatory contractual clauses and the certification requirements seem to be focused on addressing potential security risks associated with the transfer of data overseas in accordance with China’s cyber/data security law and standards.⁸⁸

(c) Security as Industrial Policy? – An Appraisal

Our review of China’s regulation of data localization and cross-border transfer provides little evidence to suggest an abuse of national security for digital industrialization. Rather, it depicts a balanced approach in China’s pursuit of development and security interests in data regulation. While data localization and cross-border transfer restrictions are widely perceived as major forms of digital protectionism,⁸⁹ China’s regulatory approach seems to be aimed at establishing a reasonable level of specificity and proportionality so that security measures target genuine risks and do not unnecessarily impede trade, investment or other business activities.

There are some facts to support this observation. For instance, China’s National Data Administration reported that by the end of 2024, the CAC completed security reviews of 285 cross-border data transfer cases, among which only 27 cases (i.e. less than 10%) failed to pass the review, mostly due to procedural issues and only a few concerning important data.⁹⁰ The CAC

⁸⁴ See above n 74, Facilitating and Standardizing Provisions, art 7.

⁸⁵ ‘《数据出境安全评估办法》 答记者问’ [Press Conference: Measures for Security Assessment of Outbound Data Transfer], Cyberspace Administration of China (7 July 2022), https://www.cac.gov.cn/2022-07/07/c_1658811536800962.htm.

⁸⁶ See above n 64, Security Assessment Measures, art 8.

⁸⁷ See above n 74, Facilitating and Standardizing Provisions, art 8.

⁸⁸ See above n 64, Standard Contract Measures, art 5; 《网络安全标准实践指南—个人信息跨境处理活动安全认证规范 V2.0》 [Cybersecurity Standards Practice Guidelines: Security Certification Specifications for Cross-Border Personal Information Processing Activities], issued by the Secretariat of the National Information Security Standardization Technical Committee on 16 December 2022, <https://www.tc260.org.cn/front/postDetail.html?id=20221216161852>, art 5.4.

⁸⁹ For an assessment of the economic costs associated with restrictions on cross-border data flow and data localization requirements, see generally OECD and WTO, *Economic Implications of Data Regulation: Balancing Openness and Trust* (OECD Publishing, Paris, 2025), https://www.wto.org/english/res_e/booksp_e/data_regulation_e.pdf.

⁹⁰ ‘国家数据局举行专题新闻发布会，介绍“关于推动数据产业高质量发展和促进企业数据资源开发利用”相关情况’ [Press Conference: Promoting High-Quality Development of the Data Industry and Facilitating the

confirmed that as of March 2025, it rejected only 7 out of 44 notified cases involving outbound transfer of important data.⁹¹ In addition, a survey conducted by the European Chamber in 2023 showed that a vast majority of European companies (70%) were able to transfer data out of China via the standard contract path without having to undertake a security review.⁹² Lately, the European Chamber's Business Confidence Survey 2025 also suggested reasonably high satisfaction with China's cross-border data transfer mechanism, with 19% of respondents expressing dissatisfaction.⁹³ This business confidence emerged from China's regulatory improvements which reduced administrative burdens for foreign business especially in free trade zones (FTZs).⁹⁴ Similarly, a recent survey conducted by the American Chamber of Commerce in China found that only 6% of responding companies have experienced difficulties in gaining approval for cross-border data transfer vital to business operations.⁹⁵

As noted, the positive developments of China's balancing act have come out of its continuous effort to streamline its data regulatory mechanisms. In March 2024, the CAC, for the first time, introduced a 'negative list' approach for FTZs, allowing those areas to develop their own catalogue of data that needs to be subject to the regulatory checks.⁹⁶ By the end of the year, Beijing, Shanghai, Tianjin, Hainan, and Zhejiang⁹⁷ released a negative list for their own FTZ to further clarify and reduce compliance requirements for cross-border data flows. To ease ongoing anxieties of foreign investors, the State Council reiterated that China is committed to facilitating "secure, orderly, and free flows of data across borders".⁹⁸ On 27 June 2025, the CAC issued the third edition of Security Review Guidelines setting out detailed document lists and procedures

Development and Utilization of Enterprises Data"], National Data Administration (31 December 2024), https://www.nda.gov.cn/sj/swdt/xwfb/1231/20241231172831486110195_pc.html.

⁹¹ '数据出境安全管理政策问答(2025年4月)'[Q&A on Security Management Policies of Data Outbound Flow (April 2025)], Cyberspace Administration of China (9 April 2025), https://www.cac.gov.cn/2025-04/09/c_1745906286623776.htm.

⁹² European Chamber, 'Flash Survey: the Impact of China's Data Regulations on European Business' (14 November 2023), <https://www.europeanchamber.com.cn/en/flash-survey-on-impact-of-china-s-data-regulations>, at 3.

⁹³ European Chamber, 'Business Confidence Survey 2025', <https://www.europeanchamber.com.cn/en/publications-business-confidence-survey>, at 1.

⁹⁴ Ibid., at 21.

⁹⁵ American Chamber of Commerce, 'AmCham China's 2025 China Business Climate Survey Report' (2025), available at: <https://www.amchamchina.org/wp-content/uploads/2025/01/AmCham-China-BCS-2025.pdf>, at 93.

⁹⁶ See above n 74, Facilitating and Standardizing Provisions, art 6. For a discussion of the negative list approach, see Guang Ma and Hong Wu, 'Cross-border Data Flow Supervision in China's Free Trade Zones: Security and Compliance Rules', (2025) *Asia Pacific Law Review*, 1-32 (Advance).

⁹⁷ '《促进和规范数据跨境流动规定》实施一周年 数据出境安全管理工作取得积极成效' [One Year after the Implementation of the Regulations on Facilitating and Standardizing Cross-Border Data Flow, Positive Results have been Achieved in the Security Management of Data Outbound Flow], Cyberspace Administration of China (21 March 2025), https://www.cac.gov.cn/2025-03/21/c_1744174598705025.htm.

⁹⁸ 《关于进一步优化外商投资环境加大吸引外商投资力度的意见》 [Opinions on Further Optimizing the Foreign Investment Environment and Attracting Foreign Investment 2023], issued by the State Council on 25 July 2023, effective on 13 August 2023, https://www.gov.cn/zhengce/content/202308/content_6898048.htm; 《扎实推进高水平对外开放更大力度吸引和利用外资行动方案》 [Action Plan on Promoting High-Level Opening Up and Attracting and Utilizing Foreign Investment 2024], issued by the General Office of State Council on 28 February 2024, effective on 19 March 2024, https://www.gov.cn/zhengce/content/202403/content_6940154.htm; 《2025年稳外资行动方案》 [Plans to Stabilize Foreign Investment 2025], issued by the State Council on 17 February 2025, https://www.gov.cn/zhengce/content/202502/content_7004409.htm.

for cross-border data transfer applications.⁹⁹ These changes were welcomed by some foreign commentators¹⁰⁰ and as noted above, gradually reduced regulatory burdens of foreign companies. While some aspects of China's regulatory framework remain problematic (as further discussed below), China seems to have committed to experimenting with new ways to soften security-related checks and requirements and liberalize data-related restrictions incrementally. China also seems willing to address concerns of foreign firms and governments about its data security regulation. For instance, Chinese authorities have consulted foreign firms in the policymaking process and have adopted some of their recommendations.¹⁰¹ The China - EU Cross-Border Data Flow Communication Mechanism, established in August 2024, is further evidence of China's willingness to collaborate with foreign governments to continuously improve the balance between its regulatory need and the need to facilitate cross-border flows of data for trade and investment.¹⁰²

China's balanced approach towards data security can be explained in at least four ways. The first relates to China's ambition to advance digital transformation which requires a market-oriented data sector to stimulate competition, innovation and growth while maintaining government interventions, including security measures, only to the extent necessary.¹⁰³ Secondly, a market-based regulatory framework is also aligned with China's long-term opening-up strategy which includes attracting foreign investment by consecutively reducing regulatory restrictions and enhancing the protection of foreign investors.¹⁰⁴ Avoiding excessive and discriminatory requirements on data localization and cross-border data flows is key to ensuring the efficacy and attractiveness of China's foreign investment policy in the digital age.¹⁰⁵ Thirdly, a balanced approach is necessary for advancing China's global strategies and engagement, particularly its continued effort to join more advanced trade or digital economy agreements such as the CPTPP and the DEPA.¹⁰⁶ These agreements and many other free trade agreements (FTAs) concluded in recent times require the parties to facilitate free flows of data across borders and minimize data localization requirements.¹⁰⁷ Using security measures for digital industrial policy would

⁹⁹ 《数据出境安全评估申报指南（第三版）》[Guidelines for Security Assessment Application for Outbound Data Transfer (Third Edition)], issued by the Cyberspace Administration of China on 27 June 2025, https://www.cac.gov.cn/2025-06/27/c_1752652339765002.htm.

¹⁰⁰ See eg. Reuters, 'China Relaxes Security Review Rules for Some Data Export' (23 March 2024), <https://www.reuters.com/technology/cybersecurity/chinas-cyberspace-regulator-issues-rules-facilitate-cross-border-data-flow-2024-03-22/>.

¹⁰¹ See eg. European Chamber, 'Business Confidence Survey 2024', <https://www.europeanchamber.com.cn/en/publications-business-confidence-survey>, at 22, fn 27.

¹⁰² The European Commission, 'EU and China launch Cross-Border Data Flow Communication Mechanism' (28 August 2024), https://policy.trade.ec.europa.eu/news/eu-and-china-launch-cross-border-data-flow-communication-mechanism-2024-08-28_en.

¹⁰³ See above n 90.

¹⁰⁴ For a review of China's foreign investment regime, see Weihuan Zhou et al., 'Technology Transfer under China's Foreign Investment Regime: Does the WTO Provide a Solution?' (2020) 54(3) *Journal of World Trade* 455, 460-69. For China's current foreign investment negative list, see https://www.gov.cn/zhengce/202409/content_6973047.htm.

¹⁰⁵ See above n 98.

¹⁰⁶ State Council, 'China to speed up accession to CPTPP', (22 March 2024), https://english.www.gov.cn/news/202403/22/content_WS65fcdcf2c6d0868f4e8e555c.html; State Council, 'China to advance efforts to join DEPA: Ministry of Commerce', (21 November 2024), https://english.www.gov.cn/news/202411/21/content_WS673f328bc6d0868f4e8ed4a9.html.

¹⁰⁷ The full text of the CPTPP is accessible here: <https://www.dfat.gov.au/trade/agreements/not-yet-in-force/tpp/Pages/tpp-text-and-associated-documents>, arts 14.11 and 14.13. The full text of the DEPA is accessible here: <https://www.mfat.govt.nz/en/trade/free-trade-agreements/free-trade-agreements-in-force/digital-economy-partnership-agreement-depa/depa-text-and-resources>, arts 4.3 and 4.4. For a detailed review of digital trade

undermine not only China's effort to join these agreements but also its global reputation more generally, as China has long positioned itself as a proponent of the rules-based trading system and an opponent to the abuse of national security.¹⁰⁸ Finally and equally importantly, China does not need to use national security as a tool to promote its digital industrialization. Instead, it remains reliant on traditional policy instruments, such as subsidies, for its digital industrial policy.¹⁰⁹

While there is no compelling evidence to suggest that China has used security-based measures to bolster its data sector, these measures have continued to attract criticisms about their (potential) asymmetric impact on foreign business. For example, a recent OECD policy paper observed that China's data localization requirements fall within the "most restrictive" category among all types of such requirements adopted by a host of economies.¹¹⁰ In this connection, some have argued that the data localization requirements have increased the cost of foreign investors in China and have supported the development of local digital infrastructure and particularly data centres.¹¹¹ It has also been reported that global automakers such as BMW, Daimler, Ford and Tesla had to set up facilities in China to store data generated by their cars locally, in order to comply with China's data localization rules.¹¹² Similarly, China's security review mechanism for cross-border data transfers, despite the positive developments discussed above, has also been subject to sustained criticisms due to certain major deficiencies in the mechanism. These deficiencies, involving ambiguities in the definitions of key terms particularly "important data" and "CIIOs", the lack of progress in developing industry-specific catalogue of important data, the need to adapt to different regulatory requirements in different regions, and non-transparency of security reviews, continue to generate administrative burdens for foreign companies and potentially benefit Chinese firms.¹¹³ These ongoing concerns are valid. While the scope of important data has been

commitments in recent FTAs, see Mira Burri and Kholofelo Kugler, 'Regulatory Autonomy in Digital Trade Agreements', (2024)27(3) *Journal of International Economic Law* 397, 400-3.

¹⁰⁸ For a discussion of China's approaches to national security under the global trading system, see above n 15, Zhou et al.

¹⁰⁹ For a discussion of China's subsidization of its high tech industries and the applicability of WTO rules, see Weihuan Zhou and Meng Fang, 'Subsidizing Technology Competition: China's Evolving Practices and International Trade Regulation', (2021)30(3) *Washington International Law Journal* 470. For policy documents, see eg. '中共中央 国务院印发《数字中国建设整体布局规划》' [A Blueprint for Developing a Digital China], issued by the Central Committee of Communist Party of China and the State Council on 27 February 2023, https://www.gov.cn/zhengce/2023-02/27/content_5743484.htm; above n 60, Guiding Opinions on Promoting the High-Quality Development of the Data Industry, art 9.2.

¹¹⁰ Chiara Del Giovane et al., 'The Nature, Evolution and Potential Implications of Data Localisation Measures', OECD Trade Policy Paper No. 278 (10 November 2023), https://www.oecd.org/en/publications/the-nature-evolution-and-potential-implications-of-data-localisation-measures_179f718a-en.html, at 8-11.

¹¹¹ Emily de la Bruyère and Nathan Picarsic, 'China's Quest for Asymmetric Dominance in Data Centers', Hinrich Foundation (25 June 2024), <https://www.hinrichfoundation.com/research/wp/tech/china-quest-for-asymmetric-dominance-in-data-centers/>, at 8.

¹¹² Yilei Sun and Tony Munroe, 'As China Plans New Rules, Global Automakers Move to Store Car Data Locally', Reuters (27 May 2021), <https://www.reuters.com/business/exclusive-china-plans-new-rules-global-automakers-move-store-car-data-locally-2021-05-27/>.

¹¹³ See eg. European Chamber, 'European Business in China Position Paper 2024/2025', https://european-chamber.com/en/publications-beijing.aliyuncs.com/upload/documents/documents/European_Business_in_China_Position_Paper_2024_2025_5.%5B1269%5D.pdf, at 30-31; European Chamber, 'Siloing and Diversification: One World, Two Systems' (9 January 2025), https://www.european-chamber.com/en/publications-archive/1272/Siloing_and_Diversification_One_World_Two_Systems, at 26.

increasingly refined especially through negative lists in the FTZs,¹¹⁴ these lists vary significantly leading to continued regulatory fragmentation. The nationwide application of this negative list approach will take time, so companies in non-FTZ areas continue to face ambiguities about the boundaries of “important data”. Moreover, although the CAC has been optimizing the workflow for security reviews, details of completed reviews remain undisclosed to the public. The combination of these outstanding issues provides wide discretion in security reviews which may be abused to benefit China’s data sector when needed.¹¹⁵ Towards this end, it is also worth noting that like many other economies, China adopts a broad definition of “national security” in its legislative system,¹¹⁶ thereby leaving considerable leeway for implementation, including for industrial policy goals. Given the global industrial policy race in the digital domain, a major way to minimise such abuse of national security as industrial policy is for all governments involved to tackle security-based data policies via concerted effort.

IV. MODERNIZING SECURITY EXCEPTIONS IN TRADE AGREEMENTS

As governments continue to liberalize data localization and cross-border data transfer requirements through trade or digital economy agreements, they have also created a range of exceptions and carve-outs for regulatory intervention on privacy, cybersecurity, and other public policy objectives. The potential policy space these exceptions may provide for data regulation has been discussed in recent studies.¹¹⁷ While we do not intend to examine all these exceptions in this paper, we note that a common feature of some major exceptions is that they do not define or constrain policy objectives governments may pursue.¹¹⁸ As such, these exceptions may be employed to justify (digital) industrial policy for developmental or strategic goals. The same argument can be made based on the WTO jurisprudence on the “public morals” exception under GATT Article XX(a). In *US – Tariff Measures (China)*, the panel endorsed the expansive interpretation of “public morals” by previous panels, holding that this exception may encompass any “standards of right and wrong” in a given society for economic and non-economic interests.¹¹⁹ In practice, WTO panels have accepted some remarkably broad policy objectives as “public morals” including “measures targeting money laundering, bridging the digital divide or promoting social inclusion, ... fraud prevention [, or theft, misappropriation and unfair competition]”.¹²⁰ Applying this broad interpretation, one may argue that the “public morals” exception also provides room for industrial policies supported by a given society to foster select sectors and create national champions for shared goals of economic development and competitiveness.

¹¹⁴ The Negative Lists for FTZs in Beijing, Shanghai, Zhejiang, Tianjin and Hainan are accessible here: https://wap.cac.gov.cn/wxzw/sjzl/sjcjfmq/A09370806phoneindex_1.htm

¹¹⁵ See eg. Raymond Yang Gao, ‘A Battle of the Big Three? – Competing Conceptualizations of Personal Data Shaping Transnational Data Flows’ (2023), 22(4) *Chinese Journal of International Law* 707, 771; above n 100, Reuters.

¹¹⁶ See generally above n 15, Zhou et al.

¹¹⁷ See generally above n 107, Burri and Kugler; Shin-Yi Peng, *International Economic Law in the Era of Datafication* (Cambridge: Cambridge University Press, 2024).

¹¹⁸ One example is the reference to “legitimate public policy objective” without providing an exhaustive list of such objectives, as further discussed below.

¹¹⁹ WTO Panel Report, *United States – Tariff Measures on Certain Goods from China*, WT/DS543/R (circulated 15 September 2020) paras. 7.131, 7.137 & 7.140.

¹²⁰ *Ibid.*, paras. 7.137 & 7.140.

Having noted the broader set of exceptions above, we turn back to our core inquiry about the nexus between national security and industrial policy by focusing on examining whether security exceptions in major trade agreements are reasonably balanced to allow room for legitimate regulation of data security while constraining abuses for protectionist or strategic economic goals.

1. Constraints (or a Lack of Constraints) of Security Exceptions under the WTO & FTAs

The security exception under the WTO was created at the conclusion of the General Agreement on Tariffs and Trade (GATT) in 1947 and remains unchanged today. As of this writing, WTO panels have interpreted and applied this exception, i.e. Article XXI of the GATT, in four disputes, all of which focused on subparagraph (b)(iii).¹²¹ Article XXI reads:¹²²

Nothing in this Agreement shall be construed

(a) to require any contracting party to furnish any information the disclosure of which it considers contrary to its essential security interests; or

(b) to prevent any contracting party from taking any action which it considers necessary for the protection of its essential security interests

(i) relating to fissionable materials or the materials from which they are derived;

(ii) relating to the traffic in arms, ammunition and implements of war and to such traffic in other goods and materials as is carried on directly or indirectly for the purpose of supplying a military establishment;

(iii) taken in time of war or other emergency in international relations; or

(c) to prevent any contracting party from taking any action in pursuance of its obligations under the United Nations Charter for the maintenance of international peace and security.

The panels' rulings on three key legal issues for determining the scope of the security exception were consistent. Firstly, whether a security measure that violates WTO rules is justifiable under the exception is not self-judging. Rather, such measures must satisfy certain legal conditions, a matter that is justiciable. In *Russia – Traffic in Transit*, the panel explained that this interpretative approach reflects the intention of the drafters to strike a balance by distinguishing “military and serious security-related conflicts from economic and trade disputes” and genuine security measures from protectionism.¹²³ In *US – Origin Marking (Hong Kong, China)*, the panel elaborated that the chapeau and the subparagraphs of Article XXI are not a “single relative clause” but set out separate legal conditions, and that the phrase “which it considers” applies to the chapeau only.¹²⁴ This means that whether a contested measure satisfies these conditions is justiciable as these conditions serve to “circumscribe (and limit) the circumstances in which the invoking Member may take action which it considers necessary for the protection of its essential security interests.”¹²⁵

¹²¹ The four panel decisions are: Panel Report, *Russia – Measures Concerning Traffic in Transit*, WT/DS512/R (adopted 26 April 2019); Panel Report, *Saudi Arabia – Measures concerning the Protection of Intellectual Property Rights*, WT/DS567/R (circulated 16 June 2020); Panel Report, *United States – Certain Measures on Steel and Aluminium Products*, WT/DS544/R (circulated 9 December 2022); Panel Report, *United States – Origin Marking Requirement*, WT/DS597/R (circulated 21 December 2022).

¹²² Where data regulation raises issues under the General Agreement on Trade in Services (GATS) or the Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPs), GATS Article XIV *bis* and TRIPs Article 73(b)(iii) provide for security exceptions identical to GATT Article XXI.

¹²³ See above n 121, Panel Report, *Russia – Traffic in Transit*, paras. 7.81, 7.98; Panel Report, *US – Steel and Aluminium Products (China)*, paras. 7.105-28.

¹²⁴ See above n 121, Panel Report, *US – Origin Marking (Hong Kong, China)*, paras. 7.48-88.

¹²⁵ *Ibid.*, paras. 7.89, 7.263.

Secondly, the scope of “emergency in international relations” is highly restrictive. The panels required “emergency” to involve situations “of the utmost gravity” leading to “a breakdown or near-breakdown” in international relations.¹²⁶ In *US – Origin Marking (Hong Kong, China)*, the panel clarified that the term “international relations” “may involve diverse matters, such as political, economic, social, or cultural exchanges.”¹²⁷ This suggests that the scope of national security may extend beyond military and defence interests to include economic, social and other interests according to the needs of individual governments. However, the panel maintained the strict interpretation of “emergency” which, in its view, must involve an exceptional state of affairs rather than mere “tensions” or “divergences”.¹²⁸ Based on the above interpretation, an “emergency in international relations” was found to exist in *Russia – Traffic in Transit* involving armed conflicts between Russia and Ukraine and in *Saudi Arabia – IPRs* involving a heightened and comprehensive diplomatic crisis between Saudi Arabia and Qatar. In contrast, in *US – Steel and Aluminium Products (China)*, the panel ruled that global steel overcapacity and its impact on the industries in the US and worldwide did not amount to an emergency in international relations, although it had provoked international concerns and joint actions to address the overcapacity.¹²⁹ In *US – Origin Marking (Hong Kong, China)*, while the enactment of the Hong Kong (HK) National Security Law provoked some economic and political tensions, the panel held that these tensions did not “meet the requisite level of gravity to constitute an emergency in international relations”.¹³⁰ In reaching this conclusion, the panel also relied on the fact that the US actions targeted HK only (i.e. not China) and only certain areas of their relations while bilateral trade and cooperation had continued in many other aspects.¹³¹

Thirdly, Article XXI(b)(iii) imposes an obligation of “good faith”, which in turn entails a minimum requirement of “plausibility” between the means and the ends.¹³² For the WTO tribunal, this requirement is needed to avoid abuse of “Article XXI as a means to circumvent” GATT obligations.¹³³ However, the evidentiary standard on “plausibility” is significantly lower than that on “emergency”, merely requiring that a contested measure is not “so remote from, or unrelated to” the security interest concerned. By doing so, the tribunal sought to develop a balanced approach to the interpretation and application of the security exception by requiring the security measures to have at least some connection to the claimed objectives.

Thus, the key constraint on policy space under the security exception pertains to the high standard imposed on what situations may amount to an “emergency” in international relations. This constraint stems from the narrowness of the treaty language itself, as intended by the drafters.¹³⁴ As such, the security exception provides little room for data security measures

¹²⁶ See above n 121, Panel Report, *Russia – Traffic in Transit*, para. 7.136; Panel Report, *Saudi Arabia – IPRs*, paras. 7.257-260; Panel Report, *US – Steel and Aluminium Products (China)*, para. 7.139; Panel Report, *US – Origin Marking (Hong Kong, China)*, paras. 7.289, 7.297-98.

¹²⁷ See above n 121, Panel Report, *US – Origin Marking (Hong Kong, China)*, para. 7.280.

¹²⁸ *Ibid.*, paras. 7.289, 7.311.

¹²⁹ See above n 121, Panel Report, *US – Steel and Aluminium Products (China)*, paras. 7.142-48.

¹³⁰ See above n 121, Panel Report, *US – Origin Marking (Hong Kong, China)*, paras. 7.323-53.

¹³¹ *Ibid.*, para. 7.354.

¹³² See above n 121, Panel Report, *Russia – Traffic in Transit*, paras. 7.132, 7.138; Panel Report, *Saudi Arabia – IPRs*, paras. 7.283-88.

¹³³ See above n 121, Panel Report, *Russia – Traffic in Transit*, para. 7.133.

¹³⁴ See Mona Pinchis-Paulsen, ‘Trade Multilateralism and U.S. National Security: the Making of the GATT Security Exceptions’, (2020)41(1) *Michigan Journal of International Law* 109 (noting that the narrow interpretation of “emergency” is consistent with the drafting record of the security exception).

(including those used for industrial policies) precisely because of the requirement that an “emergency in international relations” must be present.

In contrast with the constraint on policy space above, Article XXI(a) provides considerable flexibility for governments to refrain from “furnish[ing] any information the disclosure of which it considers contrary to its essential security interests”. The term “information” is not defined or otherwise restricted and therefore has the potential to capture data. It is true that Article XXI was crafted seven decades ago when “digital economy/trade” was completely unknown. However, WTO tribunals have long endorsed an evolutionary interpretation of existing trade rules to enable an application of the rules to address “contemporary concerns of the community of nations”.¹³⁵ The fact that all of the trade or digital economy agreements discussed below refer to “information” in their cross-border data transfer provisions is strong evidence of global concerns about data trade in the digital age. Once this threshold issue is resolved, Article XXI(a), which operates independently from Article XXI(b), is not subject to any additional legal requirements as those implicated by the subparagraphs of Article XXI(b). While it would still not permit self-judging, the only applicable condition would be the minimum requirement of plausibility as per the panel’s rulings in *Russia – Traffic in Transit* discussed above.¹³⁶ Thus, restrictions on transferring data or supplying data in other ways can be justified so long as they are not implausible as measures protective of a chosen security interest. Such a minimum degree of connection between the means and the ends is not hard to establish. In addition, Article XXI(a) does not limit the entities that provide information but instead allows *any* measures to be taken to address security risks associated with the supply of *any* information by *any* entity. Therefore, when compared with Article XXI(b)(iii), Article XXI(a) almost goes to the other extreme for a lack of any substantive constraint on data security measures, thereby leaving room for the use of such measures for a mix of security and industrial policy goals. Nevertheless, Article XXI(a) may be refined by clarifying whether the provision or supply of data also catches data localization requirements or measures limiting access to data.¹³⁷

The WTO’s security exception has been further developed in recent trade or digital economy agreements in two major aspects. One is softening the legal conditions to allow more flexibility in invoking security exceptions in general. The other is expanding the scope of the exceptions for data regulation more specifically. For example, Article 32.2 of the USMCA¹³⁸ clarified that the disclosure of information includes not only actively transferring or supplying information but also allowing access to information (paragraph 1(a)). This is arguably added to close the potential loophole under GATT Article XXI(a) noted above.¹³⁹ Paragraph 1(b) then removed the subparagraphs of GATT Article XXI(b), while maintaining the chapeau so that the phrase “it considers necessary” is no longer subject to additional legal conditions. As discussed above, while the *Russia – Traffic in Transit* decision would still impose an obligation of “good faith”, the resultant minimum requirement of plausibility can be easily satisfied. Thus, the combination of

¹³⁵ Appellate Body Report, *United States – Import Prohibition of Certain Shrimp and Shrimp Products*, WT/DS58/AB/R (adopted 6 November 1998) paras. 129-131.

¹³⁶ That is, the obligation of “good faith” stems from Article XXI as a whole rather than the sub-paragraphs. See above n 121, Panel Report, *Russia – Traffic in Transit*, paras. 7.132-33, 7.138.

¹³⁷ See above n 107, Burri and Kugler, at 420-21.

¹³⁸ The full text of the USMCA is accessible here: <https://ustr.gov/trade-agreements/free-trade-agreements/united-states-mexico-canada-agreement/agreement-between>.

¹³⁹ Ibid.

the two developments significantly reduced the rigidity of security exceptions and expanded the scope of justifiable circumstances for data security measures. These developments have also been adopted in digital economy agreements such as the DEPA.¹⁴⁰ The enhanced policy space for data security is a clear response to the more liberal approaches to data flows and localization adopted in these agreements, thereby acting as a classic “safety valve” to enable progressive liberalization or more advanced rules on data.¹⁴¹

2. *Developing Security Exceptions*

Proposals for constraining the abuse of national security to protect trade liberalization and a rules-based trading system have flourished since *Russia – Traffic in Transit*.¹⁴² Yet, detailed discussions on how best to design security exceptions for data regulation are developing slowly,¹⁴³ although some of the general proposals on security exceptions can also be useful for addressing the growing tensions between trade and data security. Below, we offer some preliminary thoughts on how to move negotiations of security exceptions for legitimate data regulation forward.

Negotiation is far better than litigation, as Professor Bacchus rightly suggested.¹⁴⁴ Litigation can provide only a certain degree of gap-filling, while at the same time carrying the risk of judicial overreach.¹⁴⁵ Negotiation is the only way to address the existing legislative deficit and modernize security exceptions in response to contemporary challenges. However, the well-known difficulties in progressing and concluding negotiations on a multilateral basis mean that alternative approaches must be taken. The WTO itself has resorted to plurilateral initiatives for negotiations on emerging issues, including digital trade, to ensure the multilateral trading system stays up to date. Via the JSI, 91 WTO members concluded “a stabilised text” in July 2024,¹⁴⁶ with a plan to seek its adoption as a plurilateral agreement under Annex 4 of the WTO Agreement.¹⁴⁷ As far as data is concerned, the JSI text does not include the more advanced rules on cross-border data

¹⁴⁰ See above n 107, DEPA, art. 15.2.

¹⁴¹ See WTO, *World Trade Report 2009: Trade Policy Commitments and Contingency Measures* (Geneva: WTO, 2009) 47. Some FTAs, such as the RCEP (art 12.17), also exclude digital trade obligations from dispute settlement, as an additional or alternative approach to safeguarding policy space. See also above n 22, Gao, ‘Data Sovereignty and Trade Agreements: Three Digital Kingdoms’, at 304-06. The full text of the RCEP is accessible here: <https://www.dfat.gov.au/trade/agreements/in-force/rcep/rcep-text>.

¹⁴² See eg. Simon Lester and Huan Zhu, ‘A Proposal for “Rebalancing” To Deal With “National Security” Trade Restrictions’, (2019)42(5) *Fordham International Law Journal* 1451; Nicolas Lamp, ‘At the Vanishing Point of Law: Rebalancing, Non-Violation Claims, and the Role of the Multilateral Trade Regime in the Trade Wars’, (2019) 22(4) *Journal of International Economic Law* 721; James Bacchus, ‘The Black Hole of National Security: Striking the Right Balance for the National Security Exception in International Trade’, CATO Policy Analysis no. 936 (9 November 2022), <https://www.cato.org/sites/cato.org/files/2022-11/policy-analysis-936.pdf>; Mona Pinchis-Paulsen, ‘Let’s Agree to Disagree: A Strategy for Trade Security’, (2022)25(4) *Journal of International Economic Law* 527; Warren Maruyama and Alan W Wolff, ‘Saving the WTO from the National Security Exception’ (Working Paper No 23-2, Peterson Institute for International Economics, May 2023), www.piie.com/publications/working-papers/2023/saving-wto-national-security-exception; Stephen Kho et al., ‘The Conundrum of the Essential Security Exception: Can the WTO Resolve the GATT Article XXI Crisis and Save the Dispute Settlement Mechanism?’, (2024)40(1) *American University International Law Review* 127.

¹⁴³ See eg. above n 14, Cohen, at 814-15.

¹⁴⁴ See above n 142, Bacchus, at 11.

¹⁴⁵ For detailed discussions of the controversies about judicial overreach by the WTO Appellate Body, see eg. Weihuan Zhou and Henry Gao, ‘“Overreaching” or “Overreacting”? Reflections on the Judicial Function and Approaches of WTO Appellate Body’, (2019)53(6) *Journal of World Trade* 951.

¹⁴⁶ WTO, ‘Joint Statement Initiative on Electronic Commerce’, INF/ECOM/87 (26 July 2024).

¹⁴⁷ WTO, ‘Information on the Agreement on Electronic Commerce’, undated, https://www.wto.org/english/tratop_e/ecom_e/information_on_agreement_ecom.pdf, at 12.

transfers and data localization adopted in the CPTPP (Articles 14.11 & 14.13), the USMCA (Articles 19.11 & 19.12), and the DEPA (Articles 4.3 & 4.4). This outcome reflects the ongoing challenges for the achievement of more liberalized data-related rules among a significantly larger group of governments.¹⁴⁸ Thus, even plurilateral negotiations will take time and need to be progressed incrementally.

Nevertheless, negotiations on data security can help facilitate data liberalization and constrain the abuse of security measures for industrial policy. To do so, the negotiations need to involve detailed discussions of major security-related issues to confine security exceptions to reasonably clear parameters. One threshold question is whether a categorization of data is required. Academic work has proposed ways for data classification,¹⁴⁹ but in addition new generation (digital) trade agreements, including those noted above, have also drawn a distinction between different types of data (e.g. government data, personal data, etc.). What is lacking in the current practice, however, is a dedicated discussion of how such data classifications would help develop exceptions to specifically address data security issues. Another threshold question is whether a classification of data holders or providers is needed,¹⁵⁰ as suggested by China's regulatory approaches relating to CIIOs discussed above. RCEP's security exceptions expanded GATT Article XXI by adding an exception for security measures taken for the protection of critical public infrastructures (Article 17.13(b)(iii)). This exception, however, is currently too broad and hence requires further negotiations to clarify the coverage of critical infrastructures, perhaps with a combination of a general list and country-specific lists to establish a commonly accepted baseline while allowing certain flexibilities for individual economies.

These threshold issues implicate a risk-based approach whereby the central issue concerns the likelihood or gravity of security risks *vis-à-vis* different types of data or data holders. Apart from China's tiered risk management mechanism discussed in Section III, Professor Peng has observed a general trend of governments "moving toward risk-based approaches to protect national security and cybersecurity."¹⁵¹ Such approaches have also been incorporated into the USMCA (Article 19.15.2) and the JSI (Article 17.3) to facilitate cooperation on cybersecurity issues. A risk-based approach should also be used to advance negotiations on data security more broadly. At a systemic level, recent studies have suggested that the so-called "targeting rule" derived from the theory of distortions remains applicable to balance trade and security interests

¹⁴⁸ Submissions by JSI participants are accessible here, https://www.wto.org/english/tratop_e/ecom_e/joint_statement_e.htm. See also Nivedita Sen, 'Understanding the Role of the WTO in International Data Flows: Taking the Liberalization or the Regulatory Autonomy Path?', (2018)21(2) *Journal of International Economic Law* 323, 339-41. In contrast, some recent digital trade agreements concluded at the bilateral level have advanced the relevant commitments. One example is the EU – Singapore Digital Trade Agreement which sets out more detailed obligations to prohibit restrictions of cross-border data flows and data localization requirements (Article 5). Nevertheless, this agreement incorporates the same security exception from Article XXI(a) (Article 30 by reference to Article 16.11 of the EU – Singapore Free Trade Agreement), thereby allowing wide policy space as discussed above. The full texts of the two agreements are accessible at: https://www.bilaterals.org/IMG/pdf/eu-singapore_text_of_the_digital_trade_agreement.pdf and https://policy.trade.ec.europa.eu/eu-trade-relationships-country-and-region/countries-and-regions/singapore/eu-singapore-agreements/texts-agreements_en.

¹⁴⁹ See above n 148, Sen, at 343-46.

¹⁵⁰ For a detailed discussion of data security challenges associated with critical infrastructure and how to refine security exceptions through a distinction between critical and noncritical infrastructure, see above n 117, Peng, *International Economic Law in the Era of Datafication*, Ch 2.

¹⁵¹ *Ibid.*, at 73.

by inquiring about whether a chosen policy instrument addresses a claimed security objective at its source.¹⁵² Accordingly, this inquiry entails two key steps: (1) identifying the policy objective(s), and (2) assessing the appropriateness of the policy instrument(s). The identification of policy objectives necessarily involves an assessment of the security risks and the level of protection that a regulating government seeks to address or achieve. Understanding such risks and objectives provides the basis for evaluating the reasonableness of the chosen means. Where the CPTPP, USMCA and DEPA further liberalized cross-border data transfer and data localization requirements, they also adopted a broad exception to allow measures “necessary to achieve a legitimate public policy objective” as long as they do not cause “arbitrary or unjustifiable discrimination” or impose restrictions on data transfers or localization “greater than are necessary to achieve the objective”.¹⁵³ Since security exceptions are separate from this exception, one may argue that data security does not fall within the ambit of the undefined “legitimate public policy objective”.¹⁵⁴ Nevertheless, the legal conditions contemplated in this exception, particularly the “necessity” and “arbitrary or unjustifiable discrimination” tests, provide a good starting point for strengthening the connection between data restrictive measures and security goals beyond a minimum requirement of plausibility. In doing so, the intention is not to restrain the freedom of governments to address any perceived security risks or pursue any chosen security goals. Rather, negotiators should focus on developing mutual understanding of such risks and goals raised by their counterparts as well as reasons for adopting certain data restrictive measures. Despite the controversies over the “necessity” and “arbitrary or unjustifiable discrimination” tests,¹⁵⁵ they have been well developed through a large volume of cases and hence are better understood than any new tests. In our view, the existing case law on these tests is largely compatible with the economic principles derived from the theory of distortions in disciplining the use of protectionist policy instruments while preserving regulatory autonomy.¹⁵⁶ Thus, these tests could be further refined and elaborated to ensure data restrictive measures target security risks and objectives as directly as feasible to minimize abuses for protectionist or other strategic goals. To alleviate concerns about the potential rigidity of the “necessity” test, governments may consider adopting a country-specific negative list of non-conforming measures to accommodate different regulatory preferences, capacity constraints and other considerations.¹⁵⁷

As negotiations take place, governments should continue to use existing committees and other venues under the WTO to discuss data security and regulatory practices to build transparency

¹⁵² See generally Mona Pinchis-Paulsen et al., ‘The National Security Exception at the WTO: Should It Just Be a Matter of When Members Can Avail of it? What About How?’, (2024)23(3) *World Trade Review* 271.

¹⁵³ See above n 107, CPTPP, art. 14.11.3 & 14.13.3; DEPA, art. 4.3.3 & 4.4.3, above n 138, USMCA, art. 19.11.2.

¹⁵⁴ This observation finds support in RCEP where the security exception is separate from the “legitimate public policy objective” exception under the clauses on data localization (art 12.14.3) and cross-border data flows (art 12.15.3).

¹⁵⁵ See eg. Magdalena Słok-Wódkowska and Joanna Mazur, ‘Between Commodification and Data Protection: Regulatory Models Governing Cross-border Information Transfers in Regional Trade Agreements’, (2024)37(1) *Leiden Journal of International Law* 111, 129-30.

¹⁵⁶ See generally Weihuan Zhou, ‘In Defense of the WTO: Why Do We Need A Multilateral Trading System’, (2020)47(1) *Legal Issues of Economic Integration* 9. This means that international trade law should focus on regulating data restrictive measures without encroaching on governments’ freedom in pursuing preferred objectives or level of protection. See also Neha Mishra, *International Trade Law and Global Data Governance: Aligning Perspectives and Practices* (Oxford: Hart Publishing, 2024) 188-89.

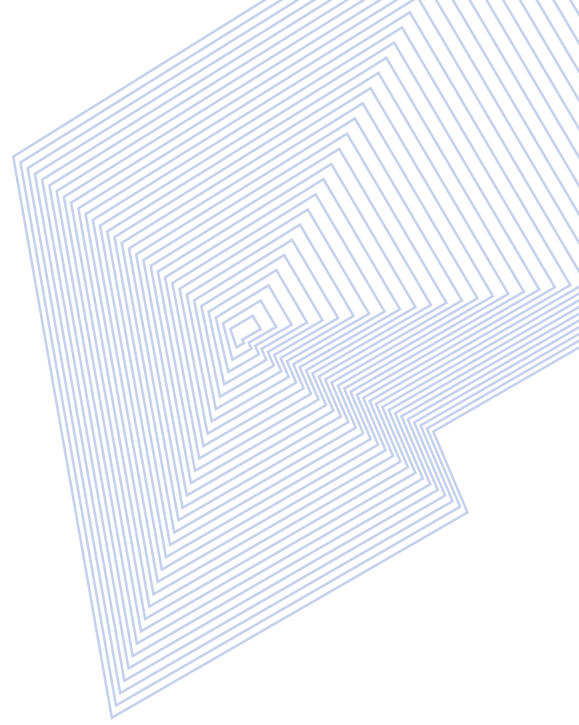
¹⁵⁷ For an example of non-conforming measures, see above n 107, CPTPP, Chapter 17-Annex IV.

and trust and manage unilateral actions and potential trade tensions via concerted effort.¹⁵⁸ Enhanced understanding of (data-related) security goals and regulatory tools, and increased trust among governments, is crucial for modernizing (data) security exceptions. The development of such exceptions would provide legal clarity on the availability of policy space as agreed by all governments involved. To the extent such development adds constraints on governments' discretions in recourse to national security, it would also provide more certainty on the enforceability of data liberalization commitments. It is from such positive spillovers of progressing negotiations on data security that progressive data liberalization may be attained and the abuse of security measures for industrial policy can be reduced.

V. CONCLUSION

Industrial policy is no longer confined to economic efficiency or competitiveness but has been increasingly motivated by a mix of economic and non-economic goals, particularly national security. This is especially so in the digital domain as governments compete for digital industrialization. While China's digital transformation has embraced ambitious industrial policies and rigorous security protection, our analysis of China's regulatory framework for data has revealed little evidence to suggest that China has (ab)used security measures to foster its data sector at the cost of foreign competitors. Rather, China's approaches to data security have arguably sought to target security risks and incrementally streamline regulatory checks and requirements. Yet, major loopholes remain in China's regulatory framework, which provide room for discretion and abuse in data security reviews. Faced with escalating geopolitical tensions and burgeoning security-based trade or investment restrictions, China's recourse to similar restrictions on security grounds is not a remote possibility. A looming data trade war in the name of national security would harm all economies as well as the rules-based global trading system they jointly built. The security exceptions under existing trade or digital economy agreements can hardly strike a proper balance between data liberalization and security. Governments should join forces to modernize these exceptions, ideally via the multilateral forum provided by the WTO for coordinated and inclusive discussions. Such discussions and negotiations would in turn promote progressive data liberalization by bringing about enhanced clarity and certainty on rebalanced rules and exceptions on data security measures. Policy space is essential in this balancing act, but a right balance cannot be achieved with security exceptions that are either overly deferential or excessively restrictive.

¹⁵⁸ For detailed discussions of how WTO committees can be used for governments to deliberate on trade-security issues, see above n 142, Pinchis-Paulsen.



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